OLIVER ZHANG

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EDUCATION

The University of Texas at Austin — May 2028
Turing Scholar Honors | B.S. Computer Science | GPA: 3.61 / 4.00
Coursework: Data Structures & Algorithms, Computer Organization & Architecture, Probability & Statistics, Linear Algebra, Discrete Math, Generalized Linear Models

EXPERIENCE

IBM – Software Engineering Intern; Research Triangle Park, NC — May 2026 – August 2026
IntelliMagic Vision ADS Agent — anomaly detection and root-cause triage for IBM Z

- Shipped a greenfield AI operations platform on IBM Z: two engineers, empty repo to 9 customers, 180+ systems in 10 weeks
- Owned the log parser and multi-agent orchestrator: extraction at 100% on the measured sample, ADK agents on Vertex AI
- Replaced a per-directory count query — 12,044 SELECTs, 0.92 TiB, 52% of BigQuery spend — with one batched prefetch

Tianjin University – Research Intern; Tianjin, China — May 2025 – August 2025

- Built a real-time computer-vision tour guide on Claude 3.5 Sonnet, sustaining 30 FPS at sub-2-second latency on CPU
- Trained and validated a TensorFlow Lite model on 5,000+ frames across lighting, angle, and location edge cases for live reliability
- Engineered a Python WebSocket back-end with a Canvas UI and Web Speech I/O, serving 3+ concurrent clients at 3x efficiency

Kyros AI (via Carnegie Mellon University faculty) – Teaching Assistant; Remote — June 2025 – December 2025

- Designed curriculum on neural networks, transformers, RAG systems and LLM fine-tuning; earned a 9/10 average student rating
- Mentored 8 students on capstone projects including custom chatbots and computer-vision applications with hands-on code reviews

PROJECTS

AI Charity Diligence Pipeline — Python, IRS 990 data, LLM tiering, Notion API — 2026 – Present

- Built the screening pipeline a student philanthropy fund runs on, cutting 1.9M+ registered nonprofits to a readable shortlist
- Routed each task to the cheapest executor that could do it: deterministic filters, then batch scoring, then reasoning models
- Calibrated the tiering against 8 prior grant winners, narrowing 1.9M organizations to 117,261 scored candidates

x86-64 Optimizing Compiler — C, x86-64 Assembly — Spring 2026

- Built an x86-64 compiler for a statically typed language; generated code ran 60× faster than the interpreted baseline
- Implemented register allocation, stack-frame management, and calling-convention compliance with verified correctness

High-Performance Matrix Multiplication — C, AVX2, pthreads, Cachegrind — Spring 2026

- Implemented 9 variants (naive, cache-blocked, recursive, SIMD, multithreaded); the pthread-blocked kernel hit 56.1 GFLOPS
- Yielded 178× speedup via ikj reordering and tiled blocking; reduced L1 miss rate 66.2% to 40.5% (Cachegrind-verified)
- Applied AVX2/FMA intrinsics (_mm256_fmadd_pd) with transposed layout; profiled GFLOPS, TSC cycles, and bandwidth

RESEARCH PUBLICATIONS

IEEE IPCCC — Carbon Impact Analysis of Large Language Models — March 2023 – November 2023
Coauthor | Texas Tech University

- Built a Python power-profiling framework measuring the carbon footprint of 5 GPT-family models across both CPU and GPU
- Found GPT-Neo 1.3B matched GPT-J 6B's quality at 73% less energy; GPUs cut inference energy 91% via a 17× speedup

HONORS & AWARDS

Citadel Securities Engineering Challenge at UT Austin — 3rd Place (January 2026) • IARIA Green 2023 Conference — Best Paper Award (August 2023)

SKILLS

Languages: C, C++, Java, Python, JavaScript, SQL, x86-64 Assembly, Bash
Core CS: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Concurrency (pthreads), Debugging & Profiling (GDB, Valgrind/Cachegrind)
Back-End & Web: REST/WebSocket APIs, client–server design, Node.js, Express, Flask, React, HTML/CSS, SQLite, Linux/Unix, Git, Docker, AWS
ML & Data: PyTorch, TensorFlow/TFLite, OpenCV, NumPy, Pandas, LangChain